

Package ‘manMetaVAR’

April 10, 2026

Title Meta-Analytic SEM for Multilevel Dynamics

Version 0.5.8

Description Research compendium for the manuscript
From Person-Specific Dynamics to Population Inference:
A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework
for Integrating Complex Multilevel Dynamic Models.
<doi:10.0000/0000000000>.

URL <https://github.com/jeksterslab/manMetaVAR>,
<https://jeksterslab.github.io/manMetaVAR/>,
<https://osf.io/uenr7/>, <https://doi.org/10.0000/0000000000>

BugReports <https://github.com/jeksterslab/manMetaVAR/issues>

License MIT + file LICENSE

Encoding UTF-8

LazyData true

Roxygen list(markdown = TRUE)

Depends R (>= 4.5.0), OpenMx (>= 2.22.11), fitVARMxID (>= 1.0.2),
metaDyn (>= 1.0.1)

Imports stats, utils, simStateSpace (>= 1.2.16), lavaan (>= 0.6-21),
ggplot2

Remotes jeksterslab/fitVARMxID, jeksterslab/metaDyn

Suggests knitr, rmarkdown, testthat

Config/roxygen2/version 7.3.3.9000

NeedsCompilation no

Author Anonymous Author [aut, cre]

Maintainer Anonymous Author <email@email.com>

Contents

Check	2
CheckFitDTVAR	3
CheckFitMetaVAR	4
CheckFitMplus	4
CheckFitNaive	5
Compress	6
FigMetricHeatmap	6
FigOverview	7
FitDTVAR	8
FitMetaVAR	8
FitMplus	9
FitNaive	10
GenData	11
manmetavar-data-methods	12
manmetavar-dtvar-methods	12
manmetavar-metavar-methods	14
manmetavar-mplus-methods	16
model	17
MplusInput	18
params	19
results	19
Sim	21
SimFitDTVAR	22
SimFitMetaVAR	23
SimFitMplus	24
SimFitNaive	25
SimFN	26
SimGenData	26
SimProj	27
Sum	27
SumFitMetaVARNormal	29
SumFitMetaVARRobust	29
SumFitMplus	30
SumFitNaive	31
Index	32

Check

Check Replication

Description

Check Replication

Usage

```
Check(taskid, repid, output_folder, naive, metavar, mplus)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
naive	Logical. If naive = TRUE, fit the naive approach.
metavar	Logical. If metavar = TRUE, fit the model using the metaDyn package.
mplus	Logical. If mplus = TRUE, fit the model using DSEM in Mplus.

Value

The output is saved as an external file in output_folder.

Author(s)

Anonymous

CheckFitDTVAR

Check Replication - FitDTVAR

Description

Check Replication - FitDTVAR

Usage

```
CheckFitDTVAR(taskid, repid, output_folder, suffix)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .

Details

This function is executed via the Check function.

Value

The output is saved as an external file in output_folder.

Author(s)

Anonymous

CheckFitMetaVAR

Check Replication - FitMetaVAR

Description

Check Replication - FitMetaVAR

Usage

```
CheckFitMetaVAR(taskid, repid, output_folder, suffix)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .

Details

This function is executed via the Check function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Anonymous

CheckFitMplus	<i>Check Replication - FitMplus</i>
---------------	-------------------------------------

Description

Check Replication - FitMplus

Usage

```
CheckFitMplus(taskid, repid, output_folder, suffix)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .

Details

This function is executed via the Check function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Anonymous

CheckFitNaive	<i>Check Replication - FitNaive</i>
---------------	-------------------------------------

Description

Check Replication - FitNaive

Usage

```
CheckFitNaive(taskid, repid, output_folder, suffix)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .

Details

This function is executed via the Check function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Anonymous

Compress	<i>Compress Replication</i>
----------	-----------------------------

Description

Compress Replication

Usage

`Compress(taskid, repid, output_folder)`

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Anonymous

FigMetricHeatmap	<i>Results Heatmap</i>
------------------	------------------------

Description

Plot results heatmap.

Usage

```
FigMetricHeatmap(results, metric_name, grey_scale = FALSE, values = FALSE)
```

Arguments

results	results data frame.
metric_name	Character string. "coverage" for coverage probability, "abs_rel_bias" for absolute value of the relative bias, "rmse" for RMSE, and "power" for statistical power.
grey_scale	Logical. If TRUE, use a grey-scale fill suitable for black-and-white printing.
values	Logical. If TRUE, display values inside tiles.

Author(s)

Anonymous

See Also

Other Figure Functions: [FigOverview\(\)](#)

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigMetricHeatmap(
  results = results,
  metric_name = "coverage"
)

## End(Not run)
```

FigOverview

Results Overview

Description

Plot results overview.

Usage

```
FigOverview(results)
```

Arguments

results results data frame.

Author(s)

Anonymous

See Also

Other Figure Functions: [FigMetricHeatmap\(\)](#)

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigOverview(results)

## End(Not run)
```

FitDTVAR

Fit the Model using the fitVARMxID Package

Description

The function fits the model using the [fitVARMxID](#) package.

Usage

```
FitDTVAR(data, ncores = NULL, seed = NULL)
```

Arguments

data R object. Output of the [GenData\(\)](#) function.
ncores Positive integer. Number of cores to use.
seed Integer. Random seed.

See Also

Other Model Fitting Functions: [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [FitNaive\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

FitMetaVAR

Multivariate Meta-Analysis using the metaDyn Package

Description

The function performs multivariate meta-analysis using the [metaDyn](#) package.

Usage

```
FitMetaVAR(fit, ncores = NULL, seed = NULL)
```

Arguments

fit	R object. Output of the FitDTVAR() function.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMplus\(\)](#), [FitNaive\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
meta <- FitMetaVAR(fit = fit, seed = seed)
summary(meta)
print(meta)
coef(meta)
```

```
vcov(meta)

## End(Not run)
```

FitMplus

Fit the Model using Mplus

Description

The function fits the model using Mplus.

Usage

```
FitMplus(
  data,
  chains = 2L,
  iter = 40000L,
  fscores = NULL,
  plot = FALSE,
  wd = ".",
  mplus_bin = NULL,
  ncores = NULL,
  seed = NULL
)
```

Arguments

data	R object. Output of the GenData() function.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.
wd	Character string. Working directory.
mplus_bin	Character string. Path to Mplus binary. If mplus_bin = NULL, the function will try to find the appropriate binary.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMetaVAR\(\)](#), [FitNaive\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitMplus(data = data)
print(fit)
summary(fit)

## End(Not run)
```

*FitNaive**Naive*

Description

The function performs the naive “fit-many-then-summarize”.

Usage

```
FitNaive(fit, seed = NULL)
```

Arguments

<code>fit</code>	R object. Output of the FitDTVAR() function.
<code>seed</code>	Integer. Random seed.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
naive <- FitNaive(fit = fit, seed = seed)
summary(naive)
print(naive)
coef(naive)
vcov(naive)

## End(Not run)
```

GenData

*Simulate Data***Description**

The function simulates data using the `simStateSpace::SimSSMIVary()` function.

Usage

```
GenData(taskid, seed = NULL)
```

Arguments

taskid	Positive integer. Task ID.
seed	Integer. Random seed.

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
print(data)
summary(data)
plot(data)

## End(Not run)
```

manmetavar-data-methods

*Methods for Objects of Class manmetavar.data***Description**

This page documents the available methods for objects of class `manmetavar.data`.

Usage

```
## S3 method for class 'manmetavar.data'
print(x, ...)

## S3 method for class 'manmetavar.data'
summary(object, ...)

## S3 method for class 'manmetavar.data'
plot(x, ...)
```

Arguments

x	Object of class <code>manmetavar.data</code> .
...	additional arguments.
object	Object of class <code>manmetavar.data</code> .

Author(s)

Anonymous

manmetavar-dtvar-methods

Methods for Objects of Class `manmetavar.dtvvar`

Description

This page documents the available methods for objects of class `manmetavar.dtvvar`.

Usage

```
## S3 method for class 'manmetavar.dtvvar'
coef(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  ncores = NULL,
  ...
)
```

```
## S3 method for class 'manmetavar.dtvvar'
vcov(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  robust = FALSE,
  ncores = NULL,
  ...
)
```

```
## S3 method for class 'manmetavar.dtvar'
print(
  x,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  digits = 4,
  ncores = NULL,
  ...
)

## S3 method for class 'manmetavar.dtvar'
summary(
  object,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  digits = 4,
  ncores = NULL,
  ...
)
```

Arguments

object	Object of class <code>manmetavar.dtvar</code> .
mu	Logical. If <code>mu = TRUE</code> , include estimates of the <code>mu</code> vector, if available. If <code>mu = FALSE</code> , exclude estimates of the <code>mu</code> vector.
alpha	Logical. If <code>alpha = TRUE</code> , include estimates of the <code>alpha</code> vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the <code>alpha</code> vector.
beta	Logical. If <code>beta = TRUE</code> , include estimates of the <code>beta</code> matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the <code>beta</code> matrix.
nu	Logical. If <code>nu = TRUE</code> , include estimates of the <code>nu</code> vector, if available. If <code>nu = FALSE</code> , exclude estimates of the <code>nu</code> vector.
psi	Logical. If <code>psi = TRUE</code> , include estimates of the <code>psi</code> matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the <code>psi</code> matrix.
theta	Logical. If <code>theta = TRUE</code> , include estimates of the <code>theta</code> matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the <code>theta</code> matrix.
ncores	Positive integer. Number of cores to use.

...	additional arguments.
robust	Logical. If TRUE, use robust (sandwich) sampling variance-covariance matrix. If FALSE, use normal theory sampling variance-covariance matrix.
x	Object of class <code>manmetavar.dtvvar</code> .
means	Logical. If <code>means = TRUE</code> , return means. Otherwise, the function returns raw estimates.
digits	Integer indicating the number of decimal places to display.

Author(s)

Anonymous

manmetavar-metavar-methods

Methods for Objects of Class `manmetavar.metavar`

Description

This page documents the available methods for objects of class `manmetavar.metavar`.

This page documents the available methods for objects of class `manmetavar.naive`.

Usage

```
## S3 method for class 'manmetavar.metavar'
coef(object, ...)

## S3 method for class 'manmetavar.metavar'
vcov(object, robust = NULL, ...)

## S3 method for class 'manmetavar.metavar'
print(x, alpha = NULL, robust = NULL, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
summary(object, alpha = NULL, robust = NULL, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
confint(object, parm = NULL, level = 0.95, robust = NULL, ...)

## S3 method for class 'manmetavar.naive'
coef(object, ...)

## S3 method for class 'manmetavar.naive'
vcov(object, ...)

## S3 method for class 'manmetavar.naive'
```

```
print(x, ...)

## S3 method for class 'manmetavar.naive'
summary(object, alpha = 0.05, digits = 4, ...)
```

Arguments

object	Object of class <code>manmetavar.naive</code> .
...	additional arguments.
robust	Logical. If TRUE, use robust (sandwich) sampling variance-covariance matrix. If FALSE, use normal theory sampling variance-covariance matrix. If NULL, the function will check object if robust standard errors are available.
x	Object of class <code>manmetavar.naive</code> .
alpha	Numeric vector. Significance level α .
digits	Integer indicating the number of decimal places to display.
parm	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
level	the confidence level required.

Author(s)

Anonymous

manmetavar-mplus-methods

Methods for Objects of Class `manmetavar.mplus`

Description

This page documents the available methods for objects of class `manmetavar.mplus`.

Usage

```
## S3 method for class 'manmetavar.mplus'
coef(object, median = TRUE, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
vcov(object, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
summary(object, alpha = 0.05, median = TRUE, digits = 4, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
confint(object, parm = NULL, level = 0.95, burnin = NULL, ...)
```



```
## S3 method for class 'manmetavar.mplus'
plot(
  x,
  what = "posterior",
  parm = NULL,
  level = 0.95,
  burnin = NULL,
  legend_loc = "topright",
  ...
)
```

Arguments

object	Object of class <code>manmetavar.mplus</code> .
median	Logical. If <code>median = TRUE</code> , return median of the posterior. If <code>median = FALSE</code> , return mean of the posterior.
burnin	Integer indicating initial samples to discard. If <code>burnin = NULL</code> , do not discard anything.
...	additional arguments.
alpha	Numeric vector. Significance level α .
digits	Integer indicating the number of decimal places to display.
parm	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
level	the confidence level required.
x	Object of class <code>manmetavar.mplus</code> .
what	Character string. If <code>what = "posterior"</code> , return posterior distribution plots. If <code>what = "trace"</code> , return trace plots.
legend_loc	Character string. Legend location.

Author(s)

Anonymous

model	<i>Model Parameters</i>
-------	-------------------------

Description

Model Parameters

Usage

`data(model)`

Format

A list with 15 elements:

k Number of variables.

mu_mu Mean of the set-point vector μ .

mu_sigma Covariance matrix of the parameter μ .

mu_sigma_l Cholesky factor of the covariance matrix of the parameter μ .

beta_mu Mean of lagged coefficients matrix β .

beta_sigma Covariance matrix of the parameter $\text{vec}(\beta)$.

beta_sigma_l Cholesky factor of the covariance matrix of the parameter $\text{vec}(\beta)$.

psi Process noise covariance matrix Ψ .

psi_l Cholesky factor of the process noise covariance matrix Ψ .

psi_d_ldl uc_d of the LDL' decomposition of the process noise covariance matrix Ψ . See [fitVARMxID::LDL\(\)](#).

psi_l_ldl s_l of the LDL' decomposition of the process noise covariance matrix Ψ . See [fitVARMxID::LDL\(\)](#).

ma_fixed Vector of fixed effects $\theta = [\mu, \text{vec}(\beta)]'$

ma_random Matrix of random effects $\theta = [\mu, \text{vec}(\beta)]'$

ma_random_d_ldl uc_d of the LDL' decomposition of the random effects. See [fitVARMxID::LDL\(\)](#).

ma_random_l_ldl s_l of the LDL' decomposition of the random effects. See [fitVARMxID::LDL\(\)](#).

Author(s)

Anonymous

MplusInput

Create Mplus Input File

Description

The function creates an Mplus input file.

Usage

```
MplusInput(
  fn_data,
  fn_posterior,
  fn_factorscores,
  chains,
  iter,
  fscores,
  plot,
  ncores = NULL,
  seed = NULL
)
```

Arguments

fn_data	Character string. Filename for data file.
fn_posterior	Character string. Filename for posterior output.
fn_factorscores	Character string. Filename for factor scores output.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [FitNaive\(\)](#)

Examples

```
cat(
  MplusInput(
    fn_data = "data.dat",
    fn_posterior = "posterior.dat",
    fn_factorscores = "factorscores.dat",
    chains = 2L,
    iter = 40000L,
    fscores = 1000L,
    plot = TRUE,
    ncores = NULL,
    seed = 42
  )
)
```

 params

Simulation Parameters

Description

Simulation Parameters

Usage

```
data(params)
```

Format

A dataframe with 12 rows and 3 columns:

taskid Simulation Task ID.

n Sample size.

time Number of measurement occasions. If NA, measurement occasions vary across IDs.

Author(s)

Anonymous

results

Simulation Results

Description

Simulation Results

Usage

`data(results)`

Format

A dataframe with 912 rows and 24 columns:

taskid Simulation Task ID.

replications Number of replications.

parnames Parameter names.

parameter Population parameter value.

method Method used to fit the data.

n Sample size.

time Number of measurement occasions.

ci Confidence/credible intervals method.

est Mean parameter estimate.

se Mean standard error.

t Mean test statistic.

p Mean p -value.

ll Mean lower limit of the 95% confidence/credible interval.

ul Mean upper limit of the 95% confidence/credible interval.

sig Proportion of statistically significant results.

zero_hit Proportion of replications where the confidence intervals contained zero.

- theta_hit** Proportion of replications where the confidence intervals contained the population parameter.
- sq_error** Mean squared error.
- bias** Bias.
- rel_bias** Relative bias.
- rmse** Root mean square error.
- coverage** Coverage probability.
- power** Statistical power.
- par_idx** Parameter index.

Author(s)

Anonymous

Sim	<i>Simulation Replication</i>
-----	-------------------------------

Description

Simulation Replication

Usage

```
Sim(  
  taskid,  
  repid,  
  output_folder,  
  overwrite,  
  integrity,  
  seed,  
  data,  
  naive,  
  metavar,  
  mplus,  
  chains,  
  iter,  
  fscores,  
  plot  
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
seed	Integer. Random seed.
data	Logical. Simulate data.
naive	Logical. If naive = TRUE, fit the naive approach.
metavar	Logical. If metavar = TRUE, fit the model using the metaDyn package.
mplus	Logical. If mplus = TRUE, fit the model using DSEM in Mplus.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.

Value

The output is saved as an external file in output_folder.

Author(s)

Anonymous

SimFitDTVAR

Simulation Replication - FitDTVAR

Description

Simulation Replication - FitDTVAR

Usage

```
SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Anonymous

SimFitMetaVAR

Simulation Replication - FitMetaVAR

Description

Simulation Replication - FitMetaVAR

Usage

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Anonymous

SimFitMplus	<i>Simulation Replication - FitMplus</i>
-------------	--

Description

Simulation Replication - FitMplus

Usage

```
SimFitMplus(  
  taskid,  
  repid,  
  output_folder,  
  seed,  
  suffix,  
  overwrite,  
  integrity,  
  chains,  
  iter,  
  fscores,  
  plot  
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
chains	Integer. Number of chains.

iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Anonymous

SimFitNaive	<i>Simulation Replication - FitNaive</i>
-------------	--

Description

Simulation Replication - FitNaive

Usage

SimFitNaive(taskid, repid, output_folder, seed, suffix, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Anonymous

SimFN

*Simulation File Name***Description**

Simulation File Name

Usage

SimFN(output_type, output_folder, suffix)

Arguments

output_type	Character string. Output type. Valid values include "data", "fit-dt-var-mx", "fit-meta-var-mx", and "fit-mplus".
output_folder	Character string. Output folder.
suffix	Character string. Output of manCTMed:::SimSuffix().

Value

Returns a character string file name with the output_folder in the OS-specific format.

SimGenData

*Simulation Replication - GenData***Description**

Simulation Replication - GenData

Usage

SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Anonymous

SimProj	<i>Simulation Project Name</i>
---------	--------------------------------

Description

Simulation Project Name

Usage

SimProj()

Value

Returns the project name as a character string.

Author(s)

Anonymous

Sum	<i>Summary</i>
-----	----------------

Description

Summary

Usage

```
Sum(  
  taskid,  
  reps,  
  output_folder,  
  overwrite,  
  integrity,  
  naive,  
  metavar_normal,  
  metavar_robust,  
  mplus,  
  ncores  
)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
naive	Logical. If naive = TRUE, summarize naive estimates.
metavar_normal	Logical. If metavar_normal = TRUE, summarize metaVAR model using normal confidence intervals.
metavar_robust	Logical. If metavar_robust = TRUE, summarize metaVAR model using robust (sandwich) confidence intervals.
mplus	Logical. If mplus = TRUE, summarize the DSEM model.
ncores	Positive integer. Number of cores to use.

Value

The output is saved as an external file in output_folder.

Author(s)

Anonymous

SumFitMetaVARNormal	<i>Summary: Normal Theory (FitMetaVAR)</i>
---------------------	--

Description

Summary: Normal Theory (FitMetaVAR)

Usage

```
SumFitMetaVARNormal(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Anonymous

SumFitMetaVARRobust	<i>Summary: Robust (FitMetaVAR)</i>
---------------------	-------------------------------------

Description

Summary: Robust (FitMetaVAR)

Usage

```
SumFitMetaVARRobust(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Anonymous

SumFitMplus	<i>Summary (FitMplus)</i>
-------------	---------------------------

Description

Summary (FitMplus)

Usage

SumFitMplus(taskid, reps, output_folder, overwrite, integrity, ncores)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Anonymous

SumFitNaive	<i>Summary: Naive</i>
-------------	-----------------------

Description

Summary: Naive

Usage

SumFitNaive(taskid, reps, output_folder, overwrite, integrity, ncores)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Anonymous

Index

- * **Compression Functions**
 - Compress, [6](#)
- * **Data Generation Functions**
 - GenData, [11](#)
- * **Figure Functions**
 - FigMetricHeatmap, [6](#)
 - FigOverview, [7](#)
- * **Model Fitting Functions**
 - FitDTVAR, [8](#)
 - FitMetaVAR, [8](#)
 - FitMplus, [9](#)
 - FitNaive, [10](#)
 - MplusInput, [18](#)
- * **check**
 - Check, [2](#)
 - CheckFitDTVAR, [3](#)
 - CheckFitMetaVAR, [4](#)
 - CheckFitMplus, [4](#)
 - CheckFitNaive, [5](#)
- * **compress**
 - Compress, [6](#)
- * **data**
 - model, [17](#)
 - params, [19](#)
 - results, [19](#)
- * **figure**
 - FigMetricHeatmap, [6](#)
 - FigOverview, [7](#)
- * **fit**
 - FitDTVAR, [8](#)
 - FitMplus, [9](#)
 - MplusInput, [18](#)
 - SimFitDTVAR, [22](#)
 - SimFitMetaVAR, [23](#)
 - SimFitMplus, [24](#)
 - SimFitNaive, [25](#)
- * **gendata**
 - GenData, [11](#)
 - SimGenData, [26](#)

- * **manCTMed**
 - SimFN, [26](#)
- * **manMetaVAR**
 - Check, [2](#)
 - CheckFitDTVAR, [3](#)
 - CheckFitMetaVAR, [4](#)
 - CheckFitMplus, [4](#)
 - CheckFitNaive, [5](#)
 - Compress, [6](#)
 - FigMetricHeatmap, [6](#)
 - FigOverview, [7](#)
 - FitDTVAR, [8](#)
 - FitMetaVAR, [8](#)
 - FitMplus, [9](#)
 - FitNaive, [10](#)
 - GenData, [11](#)
 - MplusInput, [18](#)
 - Sim, [21](#)
 - SimFitDTVAR, [22](#)
 - SimFitMetaVAR, [23](#)
 - SimFitMplus, [24](#)
 - SimFitNaive, [25](#)
 - SimGenData, [26](#)
 - SimProj, [27](#)
 - Sum, [27](#)
 - SumFitMetaVARNormal, [29](#)
 - SumFitMetaVARRobust, [29](#)
 - SumFitMplus, [30](#)
 - SumFitNaive, [31](#)
- * **meta**
 - FitMetaVAR, [8](#)
 - FitNaive, [10](#)
- * **methods**
 - manmetavar-data-methods, [12](#)
 - manmetavar-dtvar-methods, [12](#)
 - manmetavar-metavar-methods, [14](#)
 - manmetavar-mplus-methods, [16](#)
- * **parameters**
 - model, [17](#)

- params, 19
- * **results**
 - results, 19
- * **simulation**
 - Check, 2
 - CheckFitDTVAR, 3
 - CheckFitMetaVAR, 4
 - CheckFitMplus, 4
 - CheckFitNaive, 5
 - Sim, 21
 - SimFitDTVAR, 22
 - SimFitMetaVAR, 23
 - SimFitMplus, 24
 - SimFitNaive, 25
 - SimFN, 26
 - SimGenData, 26
 - SimProj, 27
 - Sum, 27
 - SumFitMetaVARNormal, 29
 - SumFitMetaVARRobust, 29
 - SumFitMplus, 30
 - SumFitNaive, 31
- * **summary**
 - Sum, 27
 - SumFitMetaVARNormal, 29
 - SumFitMetaVARRobust, 29
 - SumFitMplus, 30
 - SumFitNaive, 31
- Check, 2
- CheckFitDTVAR, 3
- CheckFitMetaVAR, 4
- CheckFitMplus, 4
- CheckFitNaive, 5
- coef.manmetavar.dtvvar
 - (manmetavar-dtvvar-methods), 12
- coef.manmetavar.metavar
 - (manmetavar-metavar-methods), 14
- coef.manmetavar.mplus
 - (manmetavar-mplus-methods), 16
- coef.manmetavar.naive
 - (manmetavar-metavar-methods), 14
- Compress, 6
- confint.manmetavar.metavar
 - (manmetavar-metavar-methods), 14
- confint.manmetavar.mplus
 - (manmetavar-mplus-methods), 16
- FigMetricHeatmap, 6
- FigMetricHeatmap(), 7
- FigOverview, 7
- FigOverview(), 7
- FitDTVAR, 8
- FitDTVAR(), 9–11, 18
- FitMetaVAR, 8
- FitMetaVAR(), 8, 10, 11, 18
- FitMplus, 9
- FitMplus(), 8, 9, 11, 18
- FitNaive, 10
- FitNaive(), 8–10, 18
- fitVARMxID, 8
- fitVARMxID::LDL(), 17
- GenData, 11
- GenData(), 8, 10
- manmetavar-data-methods, 12
- manmetavar-dtvvar-methods, 12
- manmetavar-metavar-methods, 14
- manmetavar-mplus-methods, 16
- metaDyn, 8
- model, 17
- MplusInput, 18
- MplusInput(), 8–11
- params, 19
- plot.manmetavar.data
 - (manmetavar-data-methods), 12
- plot.manmetavar.mplus
 - (manmetavar-mplus-methods), 16
- print.manmetavar.data
 - (manmetavar-data-methods), 12
- print.manmetavar.dtvvar
 - (manmetavar-dtvvar-methods), 12
- print.manmetavar.metavar
 - (manmetavar-metavar-methods), 14
- print.manmetavar.naive
 - (manmetavar-metavar-methods), 14
- results, 19
- Sim, 21
- SimFitDTVAR, 22

SimFitMetaVAR, [23](#)
SimFitMplus, [24](#)
SimFitNaive, [25](#)
SimFN, [26](#)
SimGenData, [26](#)
SimProj, [27](#)
simStateSpace::SimSSMIVary(), [11](#)
Sum, [27](#)
SumFitMetaVARNormal, [29](#)
SumFitMetaVARRobust, [29](#)
SumFitMplus, [30](#)
SumFitNaive, [31](#)
summary.manmetavar.data
 (manmetavar-data-methods), [12](#)
summary.manmetavar.dtvvar
 (manmetavar-dtvvar-methods), [12](#)
summary.manmetavar.metavar
 (manmetavar-metavar-methods),
 [14](#)
summary.manmetavar.mplus
 (manmetavar-mplus-methods), [16](#)
summary.manmetavar.naive
 (manmetavar-metavar-methods),
 [14](#)

vcov.manmetavar.dtvvar
 (manmetavar-dtvvar-methods), [12](#)
vcov.manmetavar.metavar
 (manmetavar-metavar-methods),
 [14](#)
vcov.manmetavar.mplus
 (manmetavar-mplus-methods), [16](#)
vcov.manmetavar.naive
 (manmetavar-metavar-methods),
 [14](#)